...of Ques. 5]

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G H Raisoni International Skill Tech University, Pune

(Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956)

FY B.Tech (Summer /SEM II)

ESE SUMMER 2026 (2025 Pattern)

Integral Calculus and Differential Equations (25UBSL1203)

[Time: 2.30 Hours] **[Max. Marks 60]**

Instructions to the candidates:

1) *All questions compulsory.*
2) *Figures to the right indicate full marks.*
3) ***For Que. 1,2,3 (a) is compulsory and solve (b) or (c), Que. 4 and 5 are all compulsory.***
4) *Neat diagrams must be drawn wherever necessary.*

Q.No.	*Sub Question*		*Marks*	*CO*	*BL*
Q. 1	*a)*	Trace the curve $x^2y^2 = a^2(y^2 - x^2)$	[5]	CO1	L2
	b)	Find the value of $\int_0^\infty \sqrt{x}e^{-\sqrt{x}}\,dx$	[5]	CO1	L3
		OR			
	c)	Prove that $\int_0^1 \frac{x^a - x^b}{\log x}dx = \log(\frac{a+1}{b+1})$ $a > 0, b > 1$	[5]	CO1	L3
Q. 2	*a)*	Evaluate $\int_0^1 \int_0^{1-x}(x+y)dydx$	[5]	CO1	L3
	b)	*Evaluate* $\iint y\,dxdy$ *over the region bounded by the curve* $y = x^2$ *and* $x + y = 4$	[5]	CO1	L3
		OR			
	c)	Evaluate $\int_0^1 \int_{x^2}^{2-x} xy\,dx\,dy$ by changing the order of integration	[5]	CO2	L3
Q. 3	*a)*	Evaluate $\int_c \bar{F}\,\overline{dr}$ for $\bar{F} = (2x+3)\bar{\imath} + xy\bar{\jmath} + (yz - x)\bar{k}$ along the straight line joining (0,0,0) to (3,1,1)	[5]	CO2	L3
	b)	Verify Green's theorem for $\bar{F} = x\bar{\imath} + y^2\bar{\jmath}$ over the first quadrant of the circle $x^2 + y^2 = a^2$	[5]	CO2	L3
		OR			
	c)	Evaluate $\oint[cosy\,\bar{\imath} + x(1 - siny)\bar{\jmath}]\,\overline{dr}$ over the closed curve bounded by $x^2 + y^2 = 0$, z=0	[5]	CO2	L3
Q.4	*a)*	Solve the exact differential equation $\frac{dy}{dx} = \frac{4x-2y+1}{2x-6y+2}$.	[5]	CO3	L3
	b)	Solve $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$.	[5]	CO3	L3
	c)	Solve the differential equation $(D^2 - 5D + 6)y = 3e^{5x} + 4$	[5]	CO3	L3
Q.5	*a)*	Solve $(D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$ by using method of Variation of parameter	[5]	CO4	L3
	b)	Solve $(2x+1)^2\frac{d^2y}{dx^2} - 2(2x+1)\frac{dy}{dx} - 12y = 6x$ by using Legendre's Differential equations	[5]	CO4	L3
	c)	Solve $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^5$ by using Cauchy's Differential equations.	[5]	CO4	L3